ENGN 2350 (3dasm), Brown University: formula sheet

Probability

cdf and pdf

$$P(z) = \Pr(z' \le z) = \int_{-\infty}^{z} p(z')\,dz', \qquad p(z) = \frac{dP}{dz}$$

Sum and product rules; Bayes' rule and PPD for data $\mathcal{D}$

$$p(\mathbf{y}) = \int p(\mathbf{y},\mathbf{z})\,d\mathbf{z}, \qquad p(\mathbf{y},\mathbf{z}) = p(\mathbf{y}|\mathbf{z})\,p(\mathbf{z})$$

$$p(\mathbf{z}|\mathcal{D}) = \frac{p(\mathcal{D}|\mathbf{z})\,p(\mathbf{z})}{\int p(\mathcal{D}|\mathbf{z})\,p(\mathbf{z})\,d\mathbf{z}}, \qquad p(y^*|\mathcal{D}) = \int p(y^*|\mathbf{z})\,p(\mathbf{z}|\mathcal{D})\,d\mathbf{z}$$

Moments and transformations

Expected value and covariance

$$\mathbb{E}[g(\mathbf{x})] = \int g(\mathbf{x})\,p(\mathbf{x})\,d\mathbf{x}, \qquad \mathbb{E}[\mathbf{A}\mathbf{x}+\mathbf{b}] = \mathbf{A}\,\mathbb{E}[\mathbf{x}] + \mathbf{b}$$

$$\mathrm{Cov}[\mathbf{x},\mathbf{y}] = \mathbb{E}[\mathbf{x}\mathbf{y}^T] - \mathbb{E}[\mathbf{x}]\,\mathbb{E}[\mathbf{y}]^T,$$

$$\mathrm{Cov}[\mathbf{x}] = \mathrm{Cov}[\mathbf{x},\mathbf{x}], \qquad \mathbb{V}[z] = \mathbb{E}[z^2] - \mathbb{E}[z]^2$$

$$\mathrm{Cov}[\mathbf{A}\mathbf{x}+\mathbf{b},\ \mathbf{C}\mathbf{y}+\mathbf{d}] = \mathbf{A}\,\mathrm{Cov}[\mathbf{x},\mathbf{y}]\,\mathbf{C}^T$$

$$\mathrm{Cov}[\mathbf{x}+\mathbf{y}] = \mathrm{Cov}[\mathbf{x}] + \mathrm{Cov}[\mathbf{y}] + \mathrm{Cov}[\mathbf{x},\mathbf{y}] + \mathrm{Cov}[\mathbf{y},\mathbf{x}]$$

$$\rho = \mathrm{Cov}[y,z]/\sqrt{\mathbb{V}[y]\,\mathbb{V}[z]} \in [-1,1]$$

$$\text{independent} \Rightarrow \mathbb{E}[\mathbf{x}\mathbf{y}^T] = \mathbb{E}[\mathbf{x}]\,\mathbb{E}[\mathbf{y}]^T \ \text{(uncorrelated)}$$

Law of total expectation and covariance

$$\mathbb{E}[\mathbf{y}] = \mathbb{E}\big[\mathbb{E}[\mathbf{y}|\mathbf{z}]\big], \qquad \mathrm{Cov}[\mathbf{y}] = \mathbb{E}\big[\mathrm{Cov}[\mathbf{y}|\mathbf{z}]\big] + \mathrm{Cov}\big[\mathbb{E}[\mathbf{y}|\mathbf{z}]\big]$$

Change of variables, $\mathbf{z} = \mathbf{g}(\mathbf{y})$ invertible

$$p_y(\mathbf{y}) = p_z\big(\mathbf{g}(\mathbf{y})\big)\,\big|\det \mathbf{J}_g(\mathbf{y})\big|, \qquad (\mathbf{J}_g)_{ij} = \frac{\partial g_i}{\partial y_j} \quad \Big(\text{1D: } \Big|\frac{dg}{dy}\Big|\Big)$$

Matrix identities

Woodbury

$$(\mathbf{Z} + \mathbf{U}\mathbf{W}\mathbf{V}^T)^{-1} = \mathbf{Z}^{-1} - \mathbf{Z}^{-1}\mathbf{U}(\mathbf{W}^{-1} + \mathbf{V}^T\mathbf{Z}^{-1}\mathbf{U})^{-1}\mathbf{V}^T\mathbf{Z}^{-1}$$

Derivatives and trace

$$\nabla_{\mathbf{u}}(\mathbf{u}^T\mathbf{A}\mathbf{u}) = (\mathbf{A}+\mathbf{A}^T)\,\mathbf{u}, \quad \nabla_{\mathbf{u}}(\mathbf{b}^T\mathbf{u}) = \mathbf{b}, \quad \partial \log|\mathbf{\Sigma}|/\partial\mathbf{\Sigma} = \mathbf{\Sigma}^{-1}$$

$$\partial\,\mathrm{tr}(\mathbf{\Sigma}^{-1}\mathbf{A})/\partial\mathbf{\Sigma} = -\mathbf{\Sigma}^{-1}\mathbf{A}\mathbf{\Sigma}^{-1}, \quad \mathbf{r}^T\mathbf{\Sigma}^{-1}\mathbf{r} = \mathrm{tr}(\mathbf{\Sigma}^{-1}\mathbf{r}\mathbf{r}^T)$$

Distributions and Gaussian identities

Gaussian, Laplace (variance $2/\lambda^2$ per component) and Dirac delta

$$\mathcal{N}(\mathbf{x}|\boldsymbol{\mu},\mathbf{\Sigma}) = \frac{1}{(2\pi)^{D/2}|\mathbf{\Sigma}|^{1/2}}\exp\Big[-\tfrac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^T\mathbf{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\Big]$$

$$\mathcal{L}(\mathbf{x}|\boldsymbol{\mu},\lambda) = \prod_{d=1}^{D}\frac{\lambda}{2}\,e^{-\lambda|x_d-\mu_d|}, \qquad \delta(\mathbf{A}\mathbf{x}) = \delta(\mathbf{x})/|\det\mathbf{A}|$$

Uniform on $[a,b]$

$$\mathcal{U}(x|a,b) = \tfrac{1}{b-a}, \qquad \mathbb{E}[x] = \tfrac{a+b}{2}, \qquad \mathbb{V}[x] = \tfrac{(b-a)^2}{12}$$

Gaussian identities

$$\mathbf{x}\sim\mathcal{N}(\boldsymbol{\mu},\mathbf{\Sigma}) \ \Rightarrow\ \mathbf{A}\mathbf{x}+\mathbf{b}\sim\mathcal{N}(\mathbf{A}\boldsymbol{\mu}+\mathbf{b},\ \mathbf{A}\mathbf{\Sigma}\mathbf{A}^T)$$

$$\mathcal{N}(\mathbf{x}|\mathbf{a},\mathbf{A})\,\mathcal{N}(\mathbf{x}|\mathbf{b},\mathbf{B}) = \mathcal{N}(\mathbf{a}|\mathbf{b},\ \mathbf{A}+\mathbf{B})\,\mathcal{N}(\mathbf{x}|\mathbf{c},\mathbf{C}),$$
$$\mathbf{C} = (\mathbf{A}^{-1}+\mathbf{B}^{-1})^{-1}, \qquad \mathbf{c} = \mathbf{C}\,(\mathbf{A}^{-1}\mathbf{a}+\mathbf{B}^{-1}\mathbf{b})$$

Joint Gaussian $(\mathbf{x}_a,\mathbf{x}_b)$ with blocks $\boldsymbol{\mu}_a,\boldsymbol{\mu}_b,\mathbf{\Sigma}_{aa},\mathbf{\Sigma}_{ab},\mathbf{\Sigma}_{bb}$:

$$p(\mathbf{x}_a) = \mathcal{N}(\mathbf{x}_a|\boldsymbol{\mu}_a,\mathbf{\Sigma}_{aa}), \qquad p(\mathbf{x}_a|\mathbf{x}_b) = \mathcal{N}(\mathbf{x}_a|\boldsymbol{\mu}_{a|b},\mathbf{\Sigma}_{a|b})$$

$$\boldsymbol{\mu}_{a|b} = \boldsymbol{\mu}_a + \mathbf{\Sigma}_{ab}\mathbf{\Sigma}_{bb}^{-1}(\mathbf{x}_b-\boldsymbol{\mu}_b), \quad \mathbf{\Sigma}_{a|b} = \mathbf{\Sigma}_{aa} - \mathbf{\Sigma}_{ab}\mathbf{\Sigma}_{bb}^{-1}\mathbf{\Sigma}_{ba}$$

Linear models (Bayesian linear regression)

Model with M basis functions $\boldsymbol{\phi}(\mathbf{x})$ and prior

$$p(y|\mathbf{x},\mathbf{w}) = \mathcal{N}\big(y\,|\,\mathbf{w}^T\boldsymbol{\phi}(\mathbf{x}),\ \sigma^2_{y|z}(\mathbf{x})\big), \quad p(\mathbf{w}) = \mathcal{N}(\mathbf{w}|\overset{<}{\boldsymbol{\mu}}_w,\overset{<}{\mathbf{\Sigma}}_w)$$

Posterior

$$\overset{>}{\mathbf{\Sigma}}_w = \big(\overset{<}{\mathbf{\Sigma}}_w^{-1} + \mathbf{\Phi}^T\mathbf{\Lambda}^{-1}\mathbf{\Phi}\big)^{-1}, \quad \overset{>}{\boldsymbol{\mu}}_w = \overset{>}{\mathbf{\Sigma}}_w\big(\overset{<}{\mathbf{\Sigma}}_w^{-1}\overset{<}{\boldsymbol{\mu}}_w + \mathbf{\Phi}^T\mathbf{\Lambda}^{-1}\mathbf{y}\big)$$

$\mathbf{\Phi}$: $N\times M$ with rows $\boldsymbol{\phi}(\mathbf{x}_n)^T$; $\mathbf{\Lambda} = \mathrm{diag}\big(\sigma^2_{y|z}(\mathbf{x}_n)\big)$

Posterior predictive distribution $\mathcal{N}\big(y^*|\mu^*,(\sigma^*)^2\big)$

$$\mu^* = \boldsymbol{\phi}(\mathbf{x}^*)^T\overset{>}{\boldsymbol{\mu}}_w, \qquad (\sigma^*)^2 = \sigma^2_{y|z}(\mathbf{x}^*) + \boldsymbol{\phi}(\mathbf{x}^*)^T\overset{>}{\mathbf{\Sigma}}_w\boldsymbol{\phi}(\mathbf{x}^*)$$

Marginal likelihood

$$p(\mathbf{y}) = \mathcal{N}\big(\mathbf{y}\,|\,\mathbf{\Phi}\overset{<}{\boldsymbol{\mu}}_w,\ \mathbf{\Phi}\overset{<}{\mathbf{\Sigma}}_w\mathbf{\Phi}^T + \mathbf{\Lambda}\big)$$

Point estimates: constant noise, $\mathbf{\Lambda} = \sigma^2\mathbf{I}_N$

Least Squares (Uniform prior, MLE):

$$\hat{\mathbf{w}} = (\mathbf{\Phi}^T\mathbf{\Phi})^{-1}\mathbf{\Phi}^T\mathbf{y}, \ \ \mathrm{Cov}[\hat{\mathbf{w}}] = \sigma^2(\mathbf{\Phi}^T\mathbf{\Phi})^{-1}$$
$$\hat{\sigma}^2_{\text{mle}} = \tfrac{1}{N}\mathrm{RSS}(\hat{\mathbf{w}}), \ \ \mathbb{E}[\mathrm{RSS}(\hat{\mathbf{w}})] = (N-M)\,\sigma^2$$
$$\mathrm{RSS}(\mathbf{w}) = \textstyle\sum_n\big(y_n - \mathbf{w}^T\boldsymbol{\phi}(\mathbf{x}_n)\big)^2 = ||\mathbf{y}-\mathbf{\Phi}\mathbf{w}||_2^2$$

Ridge (prior $\mathcal{N}(\mathbf{0},\overset{<}{\sigma}^2_w\mathbf{I})$ except for the bias w_0; MAP $= \overset{>}{\boldsymbol{\mu}}_w$):

$$\hat{\mathbf{w}} = \mathrm{argmin}\ \mathrm{RSS}(\mathbf{w}) + \alpha||\mathbf{w}||_2^2 = (\mathbf{\Phi}^T\mathbf{\Phi} + \alpha\mathbf{I}_0)^{-1}\mathbf{\Phi}^T\mathbf{y}, \ \ \alpha = \sigma^2/\overset{<}{\sigma}^2_w$$

Lasso (prior $\mathcal{L}(\mathbf{0},\overset{<}{\lambda}_w)$ except for the bias w_0; no closed-form MAP):

$$\hat{\mathbf{w}} = \mathrm{argmin}\ \mathrm{RSS}(\mathbf{w}) + \alpha_L||\mathbf{w}||_1, \ \ \alpha_L = 2\sigma^2\overset{<}{\lambda}_w$$

Gaussian processes and kernels

Prior over functions

$\mu_{y|z}(\cdot)\sim\mathcal{GP}\big(\overset{<}{\mu}_{y|z}(\cdot),\ k(\cdot,\cdot)\big)$: its values at any inputs are jointly Gaussian, with mean $\overset{<}{\mu}_{y|z}(\mathbf{x}_n)$ and covariance $\mathbf{K}$

Posterior predictive distribution $\mathcal{N}\big(y^*|\mu^*,(\sigma^*)^2\big)$, typically with $\overset{<}{\mu}_{y|z}=0$

$$\mu^* = \overset{<}{\mu}_{y|z}(\mathbf{x}^*) + \mathbf{k}^{*T}(\mathbf{K}+\mathbf{\Lambda})^{-1}(\mathbf{y}-\overset{<}{\boldsymbol{\mu}}_{y|z})$$

$$(\sigma^*)^2 = \sigma^2_{y|z}(\mathbf{x}^*) + k(\mathbf{x}^*,\mathbf{x}^*) - \mathbf{k}^{*T}(\mathbf{K}+\mathbf{\Lambda})^{-1}\mathbf{k}^*$$

$$K_{ij} = k(\mathbf{x}_i,\mathbf{x}_j), \qquad \mathbf{k}^{*T} = [k(\mathbf{x}_1,\mathbf{x}^*),\dots,k(\mathbf{x}_N,\mathbf{x}^*)]$$

Log marginal likelihood, $\mathbf{K}_y = \mathbf{K}+\mathbf{\Lambda}$, $\mathbf{r} = \mathbf{y} - \overset{<}{\boldsymbol{\mu}}_{y|z}$

$$\log p(\mathcal{D}|\boldsymbol{\theta}) = \underbrace{-\tfrac{1}{2}\mathbf{r}^T\mathbf{K}_y^{-1}\mathbf{r}}_{\text{data fit}}\ \underbrace{-\ \tfrac{1}{2}\log|\mathbf{K}_y|}_{\text{complexity}}\ -\ \tfrac{N}{2}\log 2\pi$$

Kernels, $r = ||\mathbf{x}-\mathbf{x}'||$

Basis	$\boldsymbol{\phi}(\mathbf{x})^T\overset{<}{\mathbf{\Sigma}}_w\boldsymbol{\phi}(\mathbf{x}')$, $\mathbf{K} = \mathbf{\Phi}\overset{<}{\mathbf{\Sigma}}_w\mathbf{\Phi}^T$, $\mathrm{rank}\,\mathbf{K} \le \min(N,M)$		
RBF	$s^2 e^{-\frac{r^2}{2l^2}}$	ARD	$s^2 e^{-\sum_i \frac{(x_i-x_i')^2}{2l_i^2}}$
Matérn 3/2	$s^2(1+\frac{\sqrt{3}\,r}{l})\,e^{-\frac{\sqrt{3}\,r}{l}}$	Periodic	$s^2 e^{-\frac{2\sin^2(\pi r/p)}{l^2}}$
Matérn 5/2	$s^2(1+\frac{\sqrt{5}\,r}{l}+\frac{5r^2}{3l^2})\,e^{-\frac{\sqrt{5}\,r}{l}}$	White	$s_2^2\,\delta_{ij}$

Valid kernels

Symmetric, $\mathbf{K}$ positive semi-definite; sums, products, $c\,k$ ($c>0$), $g(\mathbf{x})\,k\,g(\mathbf{x}')$ and $\exp(k)$ of kernels are kernels

Learned noise: the staggered scheme (Yi and Bessa, 2026)

Any model with a Gaussian observation distribution

Noise model, e.g. $\sigma^2_{y|z}(\mathbf{x}) = \exp\big(\mathbf{v}^T\boldsymbol{\psi}(\mathbf{x})\big)$; $\mathbf{\Psi}$: rows $\boldsymbol{\psi}(\mathbf{x}_n)^T$

Start from a constant noise. Repeat until converged:
1. **Mean step** (noise fixed): compute the PPD with $\mathbf{\Lambda} = \mathrm{diag}\big(\sigma^2_{y|z}(\mathbf{x}_n)\big)$
2. **Noise step** (mean fixed), with the PPD mean μ_n^* and the epistemic variance $\mathbb{V}[\mu_{y|z}(\mathbf{x}_n)]$ at each $\mathbf{x}_n$:

$$\hat{\mathbf{v}} = \underset{\mathbf{v}}{\mathrm{argmin}}\ \tfrac{1}{2}\sum_n\Big[\mathbf{v}^T\boldsymbol{\psi}(\mathbf{x}_n) + \tilde{r}_n^2\,e^{-\mathbf{v}^T\boldsymbol{\psi}(\mathbf{x}_n)}\Big]$$

$$\tilde{r}_n^2 = (y_n-\mu_n^*)^2 + \mathbb{V}[\mu_{y|z}(\mathbf{x}_n)]$$

Newton: $\mathbf{v}\leftarrow\mathbf{v} - \big(\mathbf{\Psi}^T\mathrm{diag}(e_n)\,\mathbf{\Psi}\big)^{-1}\mathbf{\Psi}^T(\mathbf{1}-\mathbf{e}), \ \ e_n = \tilde{r}_n^2\,e^{-\mathbf{v}^T\boldsymbol{\psi}(\mathbf{x}_n)}$

Predict: the PPD with the noise model at the new input, $\hat{\sigma}^2_{y|z}(\mathbf{x}^*)$

Error and model selection

Error metrics and expected error on unseen data

$$\mathrm{MSE} = \tfrac{1}{N}\textstyle\sum_n(y_n-\hat{y}_n)^2, \qquad R^2 = 1 - \frac{\sum_n(y_n-\hat{y}_n)^2}{\sum_n(y_n-\bar{y})^2}$$

$$\mathbb{E}\big[(y^*-\hat{y}^*)^2\big] = \underbrace{\sigma^2(\mathbf{x}^*)}_{\text{noise}} + \underbrace{\big(f(\mathbf{x}^*) - \mathbb{E}_{\mathcal{D}}[\hat{y}^*]\big)^2}_{\text{bias}^2} + \underbrace{\mathbb{V}_{\mathcal{D}}[\hat{y}^*]}_{\text{variance}}$$

Marginal likelihood and type-II estimates

$$p(\mathcal{D}|\boldsymbol{\theta}) = \int p(\mathcal{D}|\mathbf{z},\boldsymbol{\theta})\,p(\mathbf{z}|\boldsymbol{\theta})\,d\mathbf{z}$$

$$\hat{\boldsymbol{\theta}}_{\text{mle-II}} = \underset{\boldsymbol{\theta}}{\mathrm{argmax}}\ p(\mathcal{D}|\boldsymbol{\theta}), \qquad \hat{\boldsymbol{\theta}}_{\text{map-II}} = \underset{\boldsymbol{\theta}}{\mathrm{argmax}}\ p(\mathcal{D}|\boldsymbol{\theta})\,p(\boldsymbol{\theta})$$

Model structures

$$p(\mathcal{M}_i|\mathcal{D}) \propto p(\mathcal{D}|\mathcal{M}_i)\,p(\mathcal{M}_i),$$
$$p(\mathcal{D}|\mathcal{M}_i) = \textstyle\int p(\mathcal{D}|\boldsymbol{\theta},\mathcal{M}_i)\,p(\boldsymbol{\theta}|\mathcal{M}_i)\,d\boldsymbol{\theta}$$